CPU

File: cpu.kicad_sch

DDR3L

File: ddr3l.kicad_sch

Alimentazione

File: alimentazione.kicad_sch

Bus MIKILAB

File: bus-mikilab.kicad_sch

Storage

File: storage.kicad_sch

USB

File: usb.kicad_sch

Rete

File: rete.kicad_sch

Radio

File: radio.kicad_sch

GNSS

File: gnss.kicad_sch

Sheet: /
File: SchedaIMX6ULL.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 1/10

U1 NXP MCIMX6Y2CVM08AB (LWX 6ULL full feature, 792 MHz, −40..105 C, MAPBGA-289 14x14 mm passo 0.8). Sheet IMX6ULLCEC Rev 1.2 tabelle 90/91, confrontate con la netlist di una scheda KiCad aperta: 289/289.

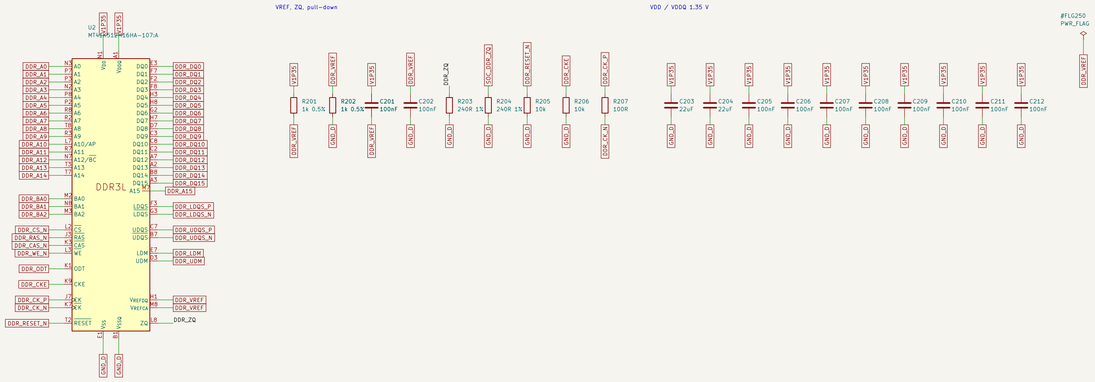
Boot: BOOT_MODE[1:0]=10 (boot interno) e BOOT_CFG tutti bassi con i pull-down interni da 100k (tabella 78) = QSPI, come SW501 tutto OFF sulla EVK NXP. Pull-up 2NP su LCD_DATA04 e LCD_DATA06 (micro50). Se QSPI e' vuota il ROM va in download seriale su USB_OTG1.

Assegnazione: ECSPI2=BUS_POWER, ECSPI1=MAIN, I2C2=bus MIKROLAB, I2C1=EEPROM MAC, UART1=console, UART3=GNSS, UART8=BT, GPT1 CAPTURE1=PPS (UART2_TX); CAPTURE2=EXTINT (UART2_RX). Elenco completo in docs/pinmap-imx6ull.md.

U1A
MCIMX6Y2CVM08AB
SOC1
LCDIF_ADDR00
LCDIF_ADDR01
LCDIF_ADDR02
LCDIF_ADDR03
LCDIF_ADDR04
LCDIF_ADDR05
LCDIF_ADDR06
LCDIF_ADDR07
LCDIF_ADDR08
LCDIF_ADDR09
LCDIF_ADDR10
LCDIF_ADDR11
LCDIF_ADDR12
LCDIF_ADDR13
LCDIF_ADDR14
LCDIF_ADDR15
LCDIF_CLK
LCDIF_CS
LCDIF_DATA00
LCDIF_DATA01
LCDIF_DATA02
LCDIF_DATA03
LCDIF_DATA04
LCDIF_DATA05
LCDIF_DATA06
LCDIF_DATA07
LCDIF_DATA08
LCDIF_DATA09
LCDIF_DATA10
LCDIF_DATA11
LCDIF_DATA12
LCDIF_DATA13
LCDIF_DATA14
LCDIF_DATA15
LCDIF_DATA16
LCDIF_DATA17
LCDIF_DATA18
LCDIF_DATA19
LCDIF_DATA20
LCDIF_DATA21
LCDIF_DATA22
LCDIF_DATA23
LCDIF_ENABLE
LCDIF_HSYNC
LCDIF_RESET
LCDIF_VSYNC
LCDIF_WR_RWN
LCDIF_RD_E
LCDIF_RS
LCDIF_CS_B
LCDIF_BUSY
LCDIF_RESET_B
LCDIF_PIXCLK
U1B
MCIMX6Y2CVM08AB
A(energia)pin1
ADC_VREFH
VREFH
ADC_VREFL
VREFL
ADC_CIC
NVCC_LCDIF
NVCC_DRAM
NVCC_DRAM_2P5
NVCC_ENET
NVCC_GPIO
NVCC_LCD
NVCC_NAND
NVCC_PLL
NVCC_SD1
NVCC_SNVS_IO
VDDA_ADC_3P3
VDD_ARM_CAP
VDD_ARM_CAP
VDD_ARM_CAP
VDD_ARM_CAP
VDD_HIGH_CAP
VDD_HIGH_CAP
VDD_SNVS_CAP
VDD_SOC_CAP
VDD_SOC_CAP
VDD_SOC_CAP
VDD_SOC_CAP
VDD_SOC_CAP
VDD_SOC_CAP
VDD_SOC_CAP
VDD_SOC_CAP
VDD_SOC_CAP
VDD_SOC_CAP
VDD_SOC_IN
VDD_USB_CAP
U1C
MCIMX6Y2CVM08AB
Masse
GND
U1D
MCIMX6Y2CVM08AB
I2C1 / BOOT_CFG
BOOT_MODE0
BOOT_MODE1
SOC_PRETTY_N
I2C1_SCL
I2C1_SDA
I2C1_DATA00
I2C1_DATA01
I2C1_DATA02
I2C1_DATA03
I2C1_DATA04
I2C1_DATA05
I2C1_DATA06
I2C1_DATA07
I2C1_DATA08
I2C1_DATA09
I2C1_DATA10
I2C1_DATA11
I2C1_DATA12
BOOT_CFG1_A
BOOT_CFG1_B
MAIN_SPI_ACTIVE_N
MAIN_SPI_N_S
MAIN_ENABLE
VBAT_ENABLE
U1E
MCIMX6Y2CVM08AB
NAND QSPI, uSDHC2, GPIO1 e SD1
NAND_ALE
NAND_CLE_B
NAND_CLE
NAND_DATA00
NAND_DATA01
NAND_DATA02
NAND_DATA03
NAND_DATA04
NAND_DATA05
NAND_DATA06
NAND_DATA07
NAND_WE_B
NAND_WP_B
NAND_READY_B
NAND_CE0_B
NAND_CE1_B
NAND_RE_B
NAND_DQS
SD1_CLK
SD1_CMD
SD1_DATA0
SD1_DATA1
SD1_DATA2
SD1_DATA3
QSPI_A_SCLK
QSPI_A_DQS
QSPI_A_SS0_B
QSPI_A_SS1_B
QSPI_A_DATA00
QSPI_A_DATA01
QSPI_A_DATA02
QSPI_A_DATA03
U1F
MCIMX6Y2CVM08AB
I2C ECSPI1/2, I2C2 e ENET2
CSI_DATA00
CSI_DATA01
CSI_DATA02
CSI_DATA03
CSI_DATA04
CSI_DATA05
CSI_DATA06
CSI_DATA07
CSI_HSYNC
CSI_MCLK
ECSPI1_SCLK
ECSPI1_MOSI
ECSPI1_MISO
ECSPI1_SS0
ENET2_RX_ER
ENET2_RX_DATA0
ENET2_RX_DATA1
ENET2_RX_EN
ENET2_TX_CLK
ENET2_TX_DATA0
ENET2_TX_DATA1
ENET2_TX_EN
MAIN_I2C_SDA
MAIN_I2C_SCL
BT_UART_RX
BT_UART_TX
U1G
MCIMX6Y2CVM08AB
ENET1, UART, GPIO1, JTAG
ENET1_RX_DATA0
ENET1_RX_DATA1
ENET1_RX_EN
ENET1_TX_CLK
ENET1_TX_DATA0
ENET1_TX_DATA1
ENET1_TX_EN
GPIO1_IO00
GPIO1_IO01
GPIO1_IO02
GPIO1_IO03
GPIO1_IO04
GPIO1_IO05
GPIO1_IO06
GPIO1_IO07
GPIO1_IO08
GPIO1_IO09
JTAG_MOD
JTAG_TCK
JTAG_TMS
JTAG_TDI
JTAG_TDO
UART1_TX_DATA
UART1_RX_DATA
UART1_CTS_B
UART1_RTS_B
UART2_TX_DATA
UART2_RX_DATA
UART2_CTS_B
UART2_RTS_B
UART3_TX_DATA
UART3_RX_DATA
UART3_CTS_B
UART3_RTS_B
UART4_TX_DATA
UART4_RX_DATA
ENET1_MDIO
ENET1_MDC
ENET1_RST_B
ENET1_TS_CLK
ENET1_TS_WAKEUP
ENET1_TS_INT
ENET1_CLK_OUT
ENET_WOL
ENET_MDIO
ENET_MDC
ENET_RST
ENET_TS_CLK
ENET_TS_WAKEUP
ENET_TS_INT
ENET_CLK_OUT
ENET_WOL
U1H
MCIMX6Y2CVM08AB
USB, clock, SNVS, sistema
CSI_CLK_B
ECSPI_CLK_B
CCM_CLK1_P
CCM_PMIC_STBY_REQ
ONOFF
PMIC_ON_REQ
RTC_XTALI
RTC_XTALO
SNVS_PMIC_ON_REQ
SNVS_TAMPER0
SNVS_TAMPER1
SNVS_TAMPER2
SNVS_TAMPER3
SNVS_TAMPER4
SNVS_TAMPER5
SNVS_TAMPER6
SNVS_TAMPER7
SNVS_TAMPER8
SNVS_TAMPER9
USB_OTG1_CHD_B
USB_OTG1_DN
USB_OTG1_DP
USB_OTG1_ID
USB_OTG1_VBUS
USB_OTG2_DN
USB_OTG2_DP
USB_OTG2_VBUS
XTALI
XTALO
VDD_SNVS_CAP
VDD_HIGH_CAP
VDD_SNVS_IN
VDD_HIGH_IN
VDD_SOC_IN
VDD_ARM_IN
NVCC_DRAM
NVCC_DRAM_2P5
NVCC_PLL
NVCC_GPIO
NVCC_SNVS_IO
NVCC_ENET
NVCC_SD1
NVCC_NAND
NVCC_LCD
NVCC_CSI
VDDA_ADC_3P3
VDDA_ADC_1P8
USB_OTG1_VBUS
USB_OTG2_VBUS
VDD_USB_CAP
VDD_SOC_IN = 1.4 V
VDD_HIGH_IN = NVCC 3.3 V (una per dominio: CSI, ENET, GPIO, LCD, NAND, SD1, UART), ADC
NVCC_DRAM 1.35 V (alto SoC)
VDD_SNVS_IN
VDD_ARM_CAP / VDD_SOC_CAP: uscite LDO interni, solo condensatori
C533 22uF
C532 1uF
C531 1uF
C534 22uF
C535 1uF
C536 1uF
C537 1uF
VDD_HIGH_CAP (a NVCC_DRAM_2P5), NVCC_PLL, VDD_SNVS_CAP, VDD_USB_CAP
C538 22uF
C539 1uF
C540 1uF
C541 1uF
C542 1uF
C543 22uF
C544 1uF
C545 1uF
C546 1uF
C547 1uF
C548 1uF
C549 22uF
C550 100nF
C551 100nF
C552 100nF
C553 100nF
C554 100nF
C555 100nF
C556 100nF
C557 100nF
C558 100nF
C559 100nF
C560 100nF
C561 100nF
C562 100nF
C563 100nF
C564 100nF
C565 100nF
C566 100nF
C567 100nF
C568 100nF
C569 100nF
C570 100nF
C571 100nF
C572 100nF
C573 100nF
C574 100nF
C575 100nF
C576 100nF
C577 100nF
Y1
24MHz
Y2
32.768kHz
Carico 24 MHz (CL da verificare sul quarzo YXC)
C578 18pF
C579 18pF
Carico RTC: circa 2xCL meno parassite (datasheet, tabella 7)
C580 12pF
C581 12pF
Boot: BOOT_MODE=10 (2NP per forzare 01 = download seriale): BOOT_CFG1[4] e [6] 0NP: JTAG_MOD a GND con 4.7k (tabella 7); BOOT_MODE rinotti a VDD_SNVS
R501 10k
R502 10k
R503 10k
R504 10k
R505 10k
R506 10k
R507 4.7k
SW1
RESET
SW2
ONOFF
J5
Console UART1
Pull-up open drain USB2_OC_N PWR_SRC_ST
R508 10k
R509 10k
IPLUG50 PWR_FLAG
IPLUG51 PWR_FLAG
IPLUG52 PWR_FLAG
IPLUG53 PWR_FLAG
IPLUG54 PWR_FLAG
IPLUG55 PWR_FLAG

U2 Micron MT41K512M16HA-107:A. 8 Gb x16 = 1 GB. FBGA-96 9x14 passo 0.8. a 1.35 V. Un solo chip punto-punto: niente VTT. Simbolo da MT41K256M16HA di KiCad con A15 su M7 (ball map JEDEC x16).

VREF a partitore 1k 0.5% con 100 nF su ogni resistenza (datasheet 60LL, tabella 3), unica per DRAM_VREF, VREFCA, VREFDQ. ZQ 240R 1% su SoC e DRAM. RESET# e CKE con 10k a GND. CK con 100R differenziale vicino alla DRAM (da confermare in simulazione).



VREF, ZQ, pull-down

VDD / VDDQ 1.35 V

Sheet: /DDR3L/		
File: ddr3l.kicad_sch		
Title:		
Size: A1	Date:	**Rev:**
KiCad E.D.A. 10.0.7		Id: 3/10

VIN_SYS 12 V dal BUS_POWER -> TPS56637 5 V (U22) -> TPS2121 U17 (IN1 bus con priorità, IN2 VBUS USB-C 1) -> 5V_SYS -> buck 3.3 V (U23). 1.35 V DDR (U26). 1.4 V VDD_SOC_IN (U27). 3.3 V SSD (U39).

Sequenza 3.3 -> 1.35 -> 1.4 V a RC sugli EN del TLV62569 (soglia EN max 1.2 V): 100k/470nF circa 21 ms, 100k/1uF circa 45 ms. POR_B dal TPS3839K33 (2.93 V, push-pull) con 1k in serie, così' SW1 puo' cortocircuitare POR_B.

VDD_SNVS: OR a diodi BAT54C tra 3V3 e CR1220 (BT1), alimenta anche V_BCKP del GNSS. Nessuna ricarica della pila. Tutti i condensatori 0603 (22uF/10V, 10uF/25V X5R).

TPS56637: EN 169k/36.1k, FB 73.2k/10k = 4.99 V, FF 20k+100pF, MODE aperto = FCCM (come schede Octavo/RK3568)

TPS2121 (come RK3568): PR1 4.49 V, OV 5.98 V, ILM 22.1k = 4–5 A, SS 100 nF

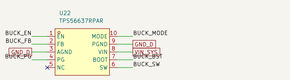
U23 3.3 V / 3 A: FB 33.2k/10k = 3.32 V, primo della sequenza

U26 1.35 V DDR: FB 150k/120k (0.6 V x 2.25), EN 100k/470nF da 3V3

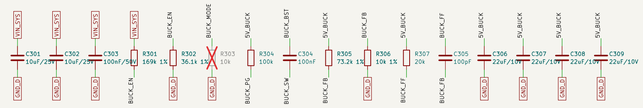
U27 1.4 V VDD_SOC_IN: FB 133k/100k = 1.398 V (min 1.375 V, 125 mV sopra VDD_ARM_CAP), EN 100k/1uF da 3V3

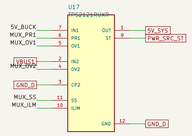
U39 3.3 V SSD, sempre acceso

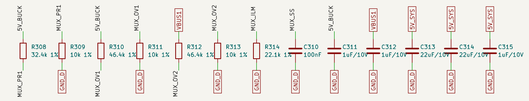
POR_B



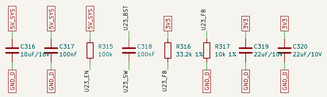




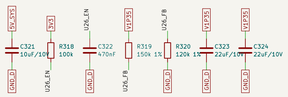






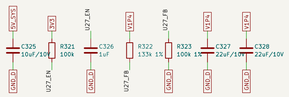














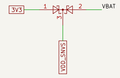














Sheet: /Alimentazione/
File: alimentazione.kicad_sch

Title:

Size: A1 | Date: | Rev:

KiCad E.D.A. 10.0.7 | Id: 4/10

J11 BUS_POWER 30 pin (Hirose FH12-30S) su ECSPI2. J12 MAIN 28 pin REV5.1 (XUNPU FPC-05F-28PH20, pin 28 = HIT#) su ECSPI1. Mappa dei connettori identica a SchedaA133/RK3568. Rail BUS_* passanti, non collegati ai rail della scheda.

Serie 33R su CLK/MOSI/MISO, 100R su ADDR/PREFIX, pull-up 10k sul CS, 4.7k su ALERT#/HIT#, 2.2k su I2C. VREF_ENABLE 10k e VBIAS_ENABLE 100k a GND. ESD TPD4E05U06DQA sulle linee logiche.

Serie SPI 33R

Serie ADDR/PREFIX 100R

Livelli di riposo

ID 000 su J_OUT (testa della catena): GND_A→GND_D ponte apribile, DNP

J11
BUS_POWER_MIKILAB

J12
BUS_MIKILAB_MAIN

U29
TPD4E05U06DQAR

U30
TPD4E05U06DQAR

U31
TPD4E05U06DQAR

U32
TPD4E05U06DQAR

U33
TPD4E05U06DQAR

U34
TPD4E05U06DQAR

Disaccoppiamento connettori

#FLG450 PWR_FLAG

#FLG451 PWR_FLAG

#FLG452 PWR_FLAG

U5 W25Q128JV su QSPI1 porta A (tabella 79, boot). J5 microSD DM3AT su uSDHC1 (sul BOTTOM). 3.3 V. CD su UART1_RTS_B (USDHC1_CD_B). U25 protegge DAT0–3: CMD e CLK non hanno ESD (un solo TPD4E in BOM, 4 linee).

U38 JMS578 (QFN-48 6x6 passo 0.4): solo USB 2.0, SuperSpeed non collegati. Datasheet Rev 1.01: 12K su REXT, 10 nF serie su SATA, 4.7 uH su LXO –> 1.2 V (VCCK/AVDDL), 30 MHz. MODE[1:0]=01, TME=0. VREG_IN e VBUS da 5V_SYS. VCCO dal suo AV330.

Niente SPI NVRAM del JMS578 (facoltativa, solo VID/PID vendor e firmware aggiornabile): usa il ROM interno. J13 solo SSD SATA: DEVSLP basso, PEDET libero. Nessun reference JMicron pubblico: da confermare sul primo prototipo.

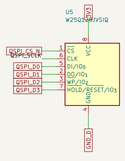
QSPI: 22R su SCLK, pull-up 10k su CS/IO2/IO3

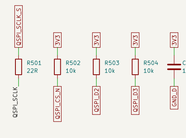
microSD: 22R su CLK, pull-up 10k

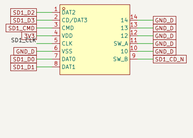
JMS578

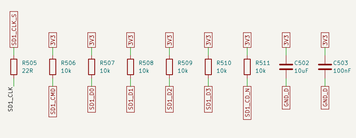
SATA: 10 nF serie (datasheet)

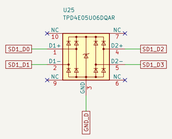
M.2

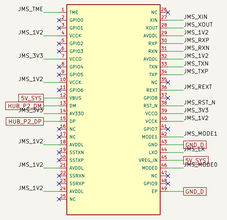




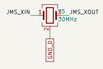






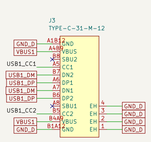


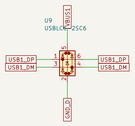




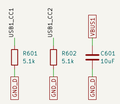
J3 USB-C 1: USB_OTG1, device/download seriale e alimentazione (VBUS1 -> IN2 del TPS2121). Sink: 5.1k su CC1/CC2. J8 USB-C 2 host: porta 1 dell'hub, VBUS dal TPS2553 (~1 A). Rp 56k. Le due USB-C sono affiancate sullo stesso bordo.

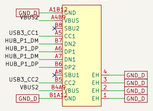
U36 FE1.1s Rev B su USB_OTG2 (3.3 V esterni su VD33F. VDD5 e pin 12/13/28 liberi come da datasheet Rev B12.4); porta 1 J8, porta 2 JMS578, porte 3–4 libere. VBUSM e BUS1 a 3.3 V (host sempre presente, self-powered). USB_OTG2_VBUS del 6ULL da 5V_SYS.

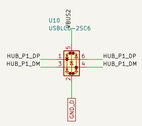




CC sink J3

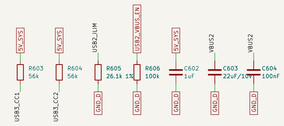


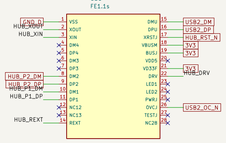






J8 host: Rp 56k. RILIM 26.1k. EN 100k a GND





Hub: REXT 2.7k, XRST/ 10k, DRV basso = hub normale

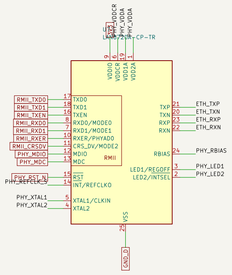
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KiCad E.D.A. 10.0.7		Id: 7/10

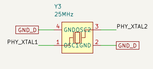
U7 LAN8720A su ENET1 RMII. Y3 25 MHz montato: il PHY genera i 50 MHz (nINTSEL basso) su REFCLKO verso ENET1_TX_CLK (REF_CLK in ingresso, SION). nINT quindi non disponibile. Strap MODE=111 (10k su RXD0/RXD1/CRS_DV). PHYADD=0.

J1 HR911105A: magnetici e LED integrati. TX/RX con 49.9R a VDDA, CT a VDDA con 22 nF. Polarita' dei LED dal disegno Hanrun: da verificare (solo indicazione, non blocca il link). U6 24AA02E48 su I2C1 con il MAC EUI-48.

PHY: terminazioni, RBIAS 12.1k, VDDCR 1uF+470pF, straps, REFCLK 22R

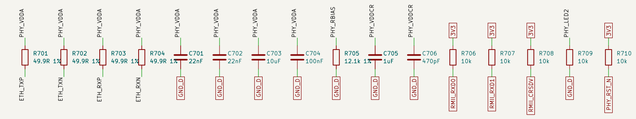
EEPROM MAC: I2C1

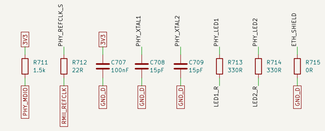


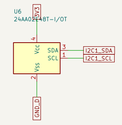


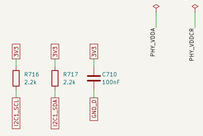












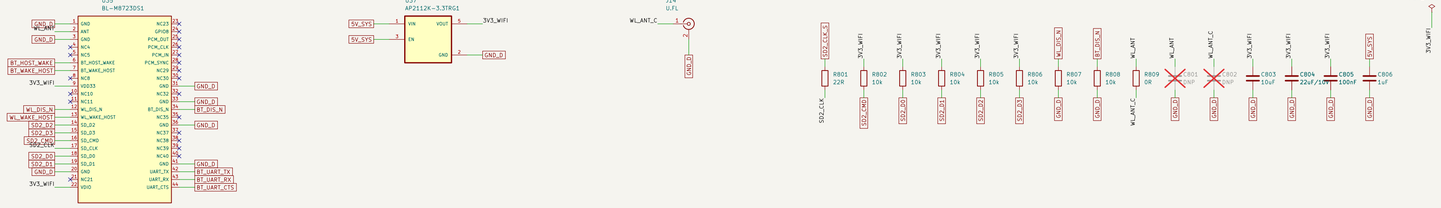


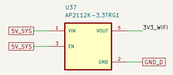
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KiCad E.D.A. 10.0.7		Id: 8/10

U35 LB-LINK BL-M8723DS1 (RTL8723DS, 12x12 mm, datasheet V1.0.1.3): Wi-Fi su uSDHC2 (pad NAND), BT su UART8 (pad ENET2). VDD33 e VOIO da U37 (3.3 V dedicati ai pilchi TX). WL_DIS# e BT_DIS# senza pull-up nel modulo: 10k a GND, li alza Linux.

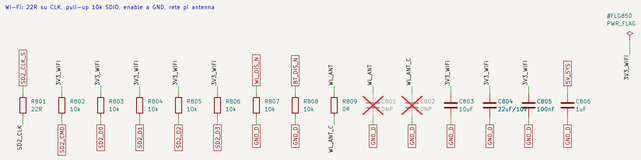
Antenna su U.FL J14 (scelta del placement: la BOM lasciava u.FL o PCB) con rete pi OR + 2 DNP da tarare. UART_CTS del modulo da UART8_RTS, il modulo non ha RTS (pin 41 = GND): H5 a 3 fili. PCM non usata: PCM_OUT libero (strap da non alzare).

Wi-Fi: 22R su CLK, pull-up 10k SDIO, enable a GND, rete pi antenna









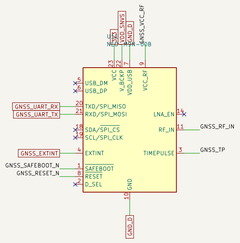
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KiCad E.D.A. 10.0.7		Id: 9/10

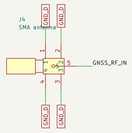
U11 u-blox NEO-M9N-00B su UART3. TIMEPULSE -> 33R -> GNSS_PPS su GPT1_CAPTURE1 (UART2_TX) e buffer U20 -> 33R -> header J7. EXTINT IN da J7 -> ESD + 100R -> buffer U19 -> GNSS_EXTINT su NEO e GPT1_CAPTURE2 (UART2_RX): evento marcato da entrambi.

J4 SMA antenna, bias 3.3 V attiva: VCC_RF -> 10R -> 27 nH (0603) -> RF_IN, 10 nF. Linea 50 ohm corta. V_BCKP da VDD_SNVS (CR1220 via BAT54C). J7: PPS OUT. EXTINT IN. GND (come scheda Tibbo).



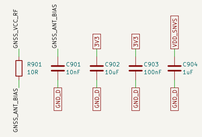




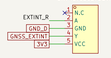




Bias antenna e alimentazione











Timing: serie e buffer